∴ Atleast one of $x \in A$ or $x \in B$ is False.

WLOG, Let $x \in A$ be False. By Defn. of negation, $\neg(x \in A)$ is True. By Defn. of Not belongs to, $x \notin A$ is True. By Defn. of characteristic function, $f_A(x) = 0$.

$\therefore f_A(x) \cdot f_B(x) = 0$

$\boxed{f_{A \cap B}(x) = f_A(x) \cdot f_B(x)}$ □

b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)$

Soln: By Defn. of characteristic function, $f_{A \cup B}(x) = \begin{cases} 1 & \text{if } x \in (A \cup B) \\ 0 & \text{if } x \notin (A \cup B) \end{cases}$

Let, $x \in (A \cup B)$. $\therefore f_{A \cup B}(x) = 1$.

By Defn. 1 from Sec-2.2, $x \in A \vee x \in B$.

The following 3 cases arise: ∵

① $x \in A$ but $x \in B$ is False.

$f_A(x) = 1$ (By defn. of characteristic function)

By Defn. of negation, $\neg(x \in B)$ is True. By defn. of not belongs to, $x \notin B$ is True. By defn. of characteristic function, $f_B(x) = 0$.

$f_A(x) \cdot f_B(x) = 0$

$\boxed{f_{A \cup B}(x) = 1 = 1 + 0 - 0 = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)}$

② $x \in B$ but $x \in A$ is False.

Similar to Case 1, we can easily prove that $\boxed{f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)}$

③ $x \in A$ and $x \in B$. By defn. of characteristic function, $f_A(x) = f_B(x) = 1$.

$\therefore f_A(x) \cdot f_B(x) = 1$.

$\boxed{f_{A \cup B}(x) = 1 = 1 + 0 = 1 + 1 - 1 = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)}$

Let, $x \notin (A \cup B)$. $\therefore f_{A \cup B}(x) = 0$.

By Defn. of not belongs to, $\neg(x \in A \cup B)$. By Defn. of negation, $\neg\neg(x \in A \cup B)$ is False. By Double-negation Law of propositions, $x \in (A \cup B)$ is False. By Defn. 1 from Sec-2.2, $x \in A \vee x \in B$ is False. By Defn. of OR, $x \in A$ is False and $x \in B$ is False. By Defn. of negation, $\neg(x \in A)$ and $\neg(x \in B)$ are both True.